We create chemistry

Safety Data Sheet
SKYTITE® C3-2.8 R RESIN

1. Identification

Product identifier used on the label

SKYTITE® C3-2.8 R RESIN

Recommended use of the chemical and restriction on use

Recommended use*: polyurethane component; industrial chemicals
Suitable for use in industrial sector: Polymers industry; chemical industry
Unsuitable for use: Uses other than recommended

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF Canada Inc.
5025 Creekbank Road
Building A, Floor 2
Mississauga, ON, L4W 0B6, CANADA

Telephone: +1 289 360-1300

Emergency telephone number

24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: (800) 454-COPE (2673)

Other means of identification

Chemical family:	resin
Synonyms:	Urethane System Resin Component

2. Hazards Identification

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

Classification of the product

Acute Tox.	4 (oral)	Acute toxicity
Skin Irrit.	2	Skin irritation
Aquatic Acute	2	Hazardous to the aquatic environment - acute

Repr.	2 (unborn child)	Reproductive toxicity
Aquatic Chronic	2	Hazardous to the aquatic environment - chronic
Eye Dam.	1	Serious eye damage
Skin Sens.	1	Skin sensitization
Carc.	2	Carcinogenicity

Label elements

Pictogram:

Signal Word:
Danger

Hazard Statement:

H318	Causes serious eye damage.
H315	Causes skin irritation.
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H351	Suspected of causing cancer.
H361	Suspected of damaging the unborn child.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P261	Avoid breathing mist or vapour or spray.
P273	Avoid release to the environment.
P202	Do not handle until all safety precautions have been read and understood.
P272	Contaminated work clothing should not be allowed out of the workplace.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or physician.
P308 + P313	IF exposed or concerned: Get medical attention.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P330	Rinse mouth.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P391	Collect spillage.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition / Information on Ingredients

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

triethyl phosphate
CAS Number: 78-40-0
Content (W/W): >= 1.0 - < 7.0%
Synonym: Phosphoric acid triethyl ester; Triethyl phosphate

cyclohexyldimethylamine
CAS Number: 98-94-2
Content (W/W): >= 0.1 - < 5.0%
Synonym: N,N-Dimethyl cyclohexanamine; Cyclohexyldimethylamine

diethylene glycol
CAS Number: 111-46-6
Content (W/W): >= 3.0 - < 10.0%
Synonym: 2,2'-Oxybisethanol

tris(2-chloro-1-methylethyl)phosphate
CAS Number: 13674-84-5
Content (W/W): >= 3.0 - < 10.0%
Synonym: 1-Chloro-2-propanol phosphate (3:1); Tris (2-chloro-1-methylethyl)phosphate

1-Propene, 1-chloro-3,3,3-trifluoro-, (1E)-
CAS Number: 102687-65-0
Content (W/W): >= 3.0 - < 10.0%
Synonym: No data available.

4-Nonylphenol, branched, ethoxylated
CAS Number: 127087-87-0
Content (W/W): >= 0.0 - < 12.0%
Synonym: 4-Nonylphenol, branched, ethoxylated

Oxirane, 2-methyl-, polymer with oxirane, ether with 2,6-bis[[bis(2-hydroxyethyl)amino]methyl]-4-branched nonylphenol
CAS Number: 940912-28-7
Content (W/W): >= 7.0 - < 20.0%
Synonym: No data available.

Ethylene oxide, polymer with 2,2'-iminodiethanol and propylene oxide
CAS Number: 34354-45-5
Content (W/W): >= 0.1 - < 5.0%
Synonym: No data available.

2,2'-dimorpholinyldiethylether
CAS Number: 6425-39-4
Content (W/W): >= 0.1 - < 5.0%
Synonym: 2,2-Dimorpholinodiethyl ether

Aromatic Amino Polyol
CAS Number: 302933-85-3

Content (W/W): >= 0.1 - < 5.0%
Synonym: (POLYMER) Formaldehyde, reaction products with branched 4-nonylphenol, diethanolamine and diisopropanolamine, ethoxylated propoxylated

4-Nonylphenol, branched, ethoxylated
CAS Number: 127087-87-0
Content (W/W): >= 1.0 - < 3.0%
Synonym: (POLYMER) Poly(oxy-1,2-ethanediyl), .alpha.-(4-nonylphenyl)-.omega.-hydroxy-,branched

Hexanoic acid, 2-ethyl-, bismuth(3+) salt
CAS Number: 67874-71-9
Content (W/W): >= 0.1 - < 3.0%
Synonym: No data available.

Bismuth, 2,2',2'',2'''-(1,2-ethanediyldinitrilo)tetrakis[ethanol] neodecanoate polypropylene glycol complexes
CAS Number: 2365490-81-7
Content (W/W): >= 0.1 - < 1.0%
Synonym: No data available.

Polyetherpolyol, halogenated
CAS Number: 68441-62-3
Content (W/W): >= 1.0 - < 5.0%
Synonym: 2-Butyne-1,4-diol polymer with (chloromethyl)oxirane, brominated, dehydrochlorinated, methoxylated

4. First-Aid Measures

Description of first aid measures

General advice:
Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

If on skin:
Wash thoroughly with soap and water

If irritation develops, seek medical attention.

If in eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

Remove contact lenses, if present.

If swallowed:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting. Immediate medical attention required.

Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far
corneal injury, Ingestion may cause gastrointestinal irritation with nausea, vomiting and diarrhea., kidney damage, skin irritation, allergic symptoms

Information on: diethylene glycol
Symptoms: Overexposure may cause:, vomiting, coma, abdominal cramps, lethargy, nausea, diarrhea, headache

Information on: tris(2-chloro-1-methylethyl)phosphate
Symptoms: Overexposure may cause:, convulsions, depression, hypoxemia, tremors

Information on: triethyl phosphate
Symptoms: Overexposure may cause:, Eye irritation, skin irritation, erythema, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

Information on: cyclohexyldimethylamine
Symptoms: Overexposure may cause:, corneal injury, skin corrosion, severe pain, coughing, respiratory disorders, dyspnea, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

Information on: 2,2'-dimorpholinyldiethylether
Symptoms: Overexposure may cause:, Eye irritation, skin irritation, erythema, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

Information on: Ethylene oxide, polymer with 2,2'-iminodiethanol and propylene oxide
Symptoms: skin irritation, corneal injury, Ingestion may cause gastrointestinal irritation with nausea, vomiting and diarrhea.

Information on: Polyetherpolyol, halogenated

Information on: Bismuth, 2,2',2'',2'''-(1,2-ethanediyldinitrilo)tetrakis[ethanol] neodecanoate polypropylene glycol complexes
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Hazards: Symptoms can appear later.

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, carbon dioxide, foam

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
No particular hazards known.

Advice for fire-fighters
Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental release measures

Further accidental release measures:
High risk of slipping due to leakage/spillage of product.

Personal precautions, protective equipment and emergency procedures
Use personal protective clothing.

Environmental precautions
Do not empty into drains. Do not discharge into the subsoil/soil.

Methods and material for containment and cleaning up
Spills should be contained, solidified, and placed in suitable containers for disposal.

7. Handling and Storage

Precautions for safe handling
Ensure thorough ventilation of stores and work areas. Avoid inhalation of mists/vapours. When using do not eat, drink or smoke. Wear suitable gloves and eye/face protection. Protect against moisture.

Protection against fire and explosion:
No special precautions necessary.

Conditions for safe storage, including any incompatibilities
Segregate from foods and animal feeds.

Suitable materials for containers: Carbon steel (Iron), High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4301 (V2)

Further information on storage conditions: From this information no suitability of the materials mentioned above for the design of installations, including containers for permanent storage, can be inferred. Special conditions apply to the selection of materials in this regard, which we can communicate on request.
Containers should be stored tightly sealed in a dry place.

Storage stability:
Storage temperature: 16 - 27 °C
Protect against moisture.
The stated storage temperature is noted for health and safety in the workplace. With regard to Quality, please refer to the product specific Technical Bulletin.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

ethylene glycol	ACGIH, US:	TWA value 25 ppm Vapor fraction ;
	ACGIH, US:	STEL value 50 ppm Vapor fraction ;
	ACGIH, US:	STEL value 10 mg/m3 Inhalable fraction, Aerosole ;

Advice on system design:
Provide local exhaust ventilation to control vapours/mists.

Personal protective equipment

Respiratory protection:
Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator as needed.

Hand protection:
Chemical resistant protective gloves, Protective glove selection must be based on the user's assessment of the workplace hazards.

Eye protection:
Wear face shield or tightly fitting safety goggles (chemical goggles) if splashing hazard exists.

Body protection:
Standard work clothes and shoes., Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures:
Avoid contact with skin and eyes. Handle in accordance with good industrial hygiene and safety practice. Wear protective clothing as necessary to prevent contact. Avoid inhalation of vapours/mists. Wash soiled clothing immediately. Do not eat, drink or use tobacco while working. Wash thoroughly after handling.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	amine-like	
Odour threshold:	No applicable information available.	
Colour:	amber	
pH value:	>= 7.0	
Freezing point:	> 0.00 °C	
Melting point:	No data available.	
Boiling point:	> 100.00 °C	
Sublimation point:	No applicable information available.	
Flash point:	> 200.00 °F	(closed cup)
Flammability:	not flammable	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling. The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	Unspecified	
Vapour pressure:	< 0.1 hPa (25 °C)	

Density:	2.45 - 2.65 lb/USg (23.88 °C)
Relative density:	No applicable information available.
Vapour density:	No applicable information available.
Partitioning coefficient n-octanol/water (log Pow):	Unspecified
Self-ignition temperature:	not self-igniting
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
Viscosity, dynamic:	400.000 mPa.s (22.00 °C)
Viscosity, kinematic:	not determined
Solubility in water:	slightly soluble
Solubility (quantitative):	No applicable information available.
Solubility (qualitative):	No applicable information available.
Molar mass:	No data available.
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.
Other Information:	If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:
No corrosive effect on metal.

Oxidizing properties:
Not an oxidizer.

Chemical stability
The product is chemically stable.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
Temperature: < 0 degrees Celsius

Incompatible materials
acids, oxidizing agents, isocyanates

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: carbon monoxide, carbon dioxide, nitrogen oxide, hydrogen cyanide

Thermal decomposition:
No decomposition if stored and handled as prescribed/indicated.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Oral
Type of value: ATE
Value: 1,980 mg/kg

Inhalation
No applicable information available.

Dermal
No applicable information available.

Assessment other acute effects
Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Origin of data: expert judgement

Irritation / corrosion
Assessment of irritating effects: May cause severe damage to the eyes. Skin contact causes irritation.

Information on: triethyl phosphate
Assessment of irritating effects: Not irritating to the skin. Eye contact causes irritation.

Information on: 2,2'-dimorpholinyldiethylether
Assessment of irritating effects: Eye contact causes irritation. Skin contact causes slight irritation.

Information on: Oxirane, 2-methyl-, polymer with oxirane, ether with 2,6-bis[[bis(2-hydroxyethyl)amino]methyl]-4-branched nonylphenol
Assessment of irritating effects: Skin contact causes irritation. May cause severe damage to the eyes.

Information on: cyclohexyldimethylamine
Assessment of irritating effects: Corrosive! Damages skin and eyes.

Information on: Bismuth, 2,2',2'',2'''-(1,2-ethanediyldinitrilo)tetrakis[ethanol] neodecanoate polypropylene glycol complexes

Information on: 4-Nonylphenol, branched, ethoxylated

Information on: Ethylene oxide, polymer with 2,2'-iminodiethanol and propylene oxide
Assessment of irritating effects: Skin contact causes irritation. May cause severe damage to the eyes.

Information on: Aromatic Amino Polyol
Assessment of irritating effects: May cause severe damage to the eyes.

Sensitization
Assessment of sensitization: Sensitization after skin contact possible.

Information on: Bismuth, 2,2',2'',2'''-(1,2-ethanediyldinitrilo)tetrakis[ethanol] neodecanoate polypropylene glycol complexes
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Aspiration Hazard
No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Repeated oral uptake of the substance did not cause substance-related effects. Repeated inhalative uptake of the substance did not cause substance-related effects. Repeated dermal uptake of the substance did not cause substance-related effects.

Information on: diethylene glycol
Assessment of repeated dose toxicity: The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies. The substance may cause damage to the liver after repeated ingestion of high doses, as shown in animal studies. The effects were only observed at doses/concentrations not relevant for classification and/or practical use conditions. These effects are not relevant to humans at occupational levels of exposure.

Genetic toxicity
Assessment of mutagenicity: The chemical structure does not suggest a specific alert for such an effect. No applicable information available.

Carcinogenicity
Assessment of carcinogenicity: Indication of possible carcinogenic effect in animal tests.

Information on: tris(2-chloro-1-methylethyl)phosphate
Assessment of carcinogenicity: Indication of possible carcinogenic effect in animal tests.

Reproductive toxicity
Assessment of reproduction toxicity: The chemical structure does not suggest a specific alert for such an effect. No applicable information available.

Teratogenicity

Information on: ethylene glycol
Assessment of teratogenicity: Developmental toxicity was observed after oral ingestion of high doses in studies with rats and mice, but this effect was not seen in a study with rabbits. Mechanistic studies show that the rabbit is the relevant species for the classification for human health. As such, and since ethylene glycol is not a developmental toxicant in the rabbit, no classification is warranted.

Information on: Hexanoic acid, 2-ethyl-, bismuth(3+) salt

Assessment of teratogenicity: Indications of possible developmental toxicity/teratogenicity were seen in animal studies. The product has not been tested. The statement has been derived from the properties of the individual components.

Other Information
The product has not been tested. The statement has been derived from the properties of the individual components.

Medical conditions aggravated by overexposure
Individuals with allergic history or pre-existing dermatitis should use extra precautions when handling this product.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Acutely toxic for aquatic organisms. Acutely harmful for aquatic organisms.

Toxicity to fish

Information on: tris(2-chloro-1-methylethyl)phosphate
LC50 (96 h) 51 mg/l, Pimephales promelas (OECD Guideline 203, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: cyclohexyldimethylamine
LC50 (96 h) 28 mg/l, Oncorhynchus mykiss (OECD 203; ISO 7346; 84/449/EWG, C.1, static)

Information on: 1-Propene, 1-chloro-3,3,3-trifluoro-, (1E)-
LC50 (96 h) 38 mg/l, Oncorhynchus mykiss (OECD 203; ISO 7346; 92/69/EWG, C.1, static)

Information on: Aromatic Amino Polyol
LC50 (96 h) 1.07 mg/l (Flow through.)

Aquatic invertebrates

Information on: tris(2-chloro-1-methylethyl)phosphate
EC50 (48 h) 131 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: cyclohexyldimethylamine
LC50 (48 h) 75 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-Propene, 1-chloro-3,3,3-trifluoro-, (1E)-
EC50 (48 h) 82 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Information on: Aromatic Amino Polyol
EC50 (48 h) 8.35 mg/l (Flow through.)

Aquatic plants

Information on: tris(2-chloro-1-methylethyl)phosphate
EC50 (72 h) 82 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Nominal concentration.
No observed effect concentration (72 h) 42 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Nominal concentration.

Information on: cyclohexyldimethylamine
EC50 (72 h) 3.5 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Nominal values (confirmed by concentration control analytics)
EC10 (72 h) 0.6 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Nominal values (confirmed by concentration control analytics)

Information on: 1-Propene, 1-chloro-3,3,3-trifluoro-, (1E)-
EC50 (72 h) > 215 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
No observed effect concentration (72 h) 115 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)

Information on: Aromatic Amino Polyol
No observed effect concentration (96 h) 0.0986 mg/l (static)
EC50 (96 h) 0.59 mg/l (static)

Chronic toxicity to aquatic invertebrates

Information on: tris(2-chloro-1-methylethyl)phosphate
No observed effect concentration (21 d) 32 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)
Nominal concentration.

Information on: cyclohexyldimethylamine
No observed effect concentration (21 d) 0.754 mg/l, Daphnia magna (other, static)

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: tris(2-chloro-1-methylethyl)phosphate
DIN EN ISO 8192 aquatic
activated sludge/EC50 (3 h): 784 mg/l
Nominal concentration.

Information on: cyclohexyldimethylamine
DIN 38412 Part 8 static
bacterium/EC10 (17 h): 137 mg/l
Nominal concentration.

Persistence and degradability

Assessment biodegradation and elimination (H2O)
Poorly biodegradable.

Elimination information

Poorly biodegradable.

Bioaccumulative potential

Assessment bioaccumulation potential
Does not significantly accumulate in organisms.

Mobility in soil

Assessment transport between environmental compartments
Adsorption to solid soil phase is not expected.

Additional information

Adsorbable organically-bound halogen(AOX):
This product contains no organically-bound halogen.

Other ecotoxicological advice:
Do not allow to enter soil, waterways or waste water channels. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

13. Disposal considerations

Waste disposal of substance:
Incinerate or dispose of in a licensed facility. Do not discharge substance/product into sewer system.

Container disposal:
Do not reuse empty containers. Under no circumstances should empty drums be burned or cut open with gas or electric torch as toxic decomposition products may be liberated. Steel drums must be emptied and can be sent to a licensed drum reconditioner for reuse, a scrap metal dealer or an approved landfill. Do not attempt to refill or clean containers since residue is difficult to remove.

14. Transport Information

Land transport
TDG

Not classified as a dangerous good under transport regulations

Sea transport
IMDG

Not classified as a dangerous good under transport regulations

Air transport
IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:
Chemical DSL, CA released; restriction on quantity / not listed

NFPA Hazard codes:
Health: 3 Fire: 1 Reactivity: 1 Special:

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2024/08/07

We support worldwide Responsible Care® initiatives. We value the health and safety of our employees, customers, suppliers and neighbors, and the protection of the environment. Our commitment to Responsible Care is integral to conducting our business and operating our facilities in a safe and environmentally responsible fashion, supporting our customers and suppliers in ensuring the safe and environmentally sound handling of our products, and minimizing the impact of our operations on society and the environment during production, storage, transport, use and disposal of our products.

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